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3 (*we list all four examples in one .wl file for the convenience of demonstratic
4 (*However, we recommend each example in one file.*)

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```

In[103]:= (*using same way of loading as AMflow and Blade*)
8 current = If[$FrontEnd===Null,$InputFileName,NotebookFileName[]]//DirectoryName
9 Get[current<>"../../" <> "BaikovAll.wl"];

```

BaikovAll: a package for generating Baikov representations for a given Feynman integral family
Version: 1.1.0
If you come across any problem, please contact phamapku14@gmail.com

Outer-Massive double box

```

In[49]:= dlist={
17     PD[k1,0,-m2],
18     PD[k1,-p1,-m2],
19     PD[k1,-p1-p2,-m2],
20     PD[k1-k2,0,0],
21     PD[k2,-p1-p2,-m2],
22     PD[k2,-p1-p2-p3,-m2],
23     PD[k2,0,-m2],
24     PD[k2,-p1,-m2],
25     PD[k1,-p1-p2-p3,-m2]
26 };(*input of propagators and independent scalar products*)
27 conservation={p4→-p1-p2-p3};(*if dependent variables does not appear, this can
28 kinematics={
29     SProd[p1]→0,
30     SProd[p2]→0,
31     SProd[p3]→0,
32     SProd[p4]→0,
33     SProd[p1+p2]→s,
34     SProd[p2+p3]→t
35 };(*define independent scalar products, SProd[a]=SProd[a,a]*)

```

```

In[52]:= kinerep=GetIndExtSProd[kinematics,conservation];(*represent all scalar products
39 krep=Join[BaikovTrans[dlist,kinerep][[2]],kinerep];(*a replacement rule for all s

```

```

In[14]:= stdrep=BaikovRep[dlist,{k1,k2},{p1,p2,p3},Abst→True];(*the standard representa
43 resultmat=AllSectorBaikovMat[stdrep,krep,Exc→{}];(*derive all representations

```

The global variable \$LoopNum has been set to 2
The global variable \$BvNum has been set to 9

```

In[17]:= Export[current<>"db_rep.wxf",resultmat,PerformanceGoal→"Size"];

```

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49 (*we can now explore some function in this family*)

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```

In[18]:= (*1. Get the loop by loop representation from the whole representations*)
53 (*now we need to specify the sector we want, {1,2,3,4,5,6,7} specifies which pr
54 (*two independent loop-by-loop representation will be extracted*)
55 toprep=GetBaikovMatRep[resultmat,{1,2,3,4,5,6,7}]

```

```

Out[18]= { { {x9}, { { {G[{p1, p2, p3}, {p1, p2, p3}], ε}, {-G[{k2, p1, p2}, {k2, p1, p2}],
-1 +  $\frac{1}{2} (2 + 2 \epsilon)$  }}, {G[{k2, p1, p2, p3}, {k2, p1, p2, p3}],  $\frac{1}{2} + \frac{1}{2} (-2 - 2 \epsilon)$  }},
{G[{k1, k2, p1, p2}, {k1, k2, p1, p2}],  $\frac{1}{2} + \frac{1}{2} (-2 - 2 \epsilon)$  }},  $\frac{1}{64 \pi^3 \text{Gamma}[\frac{1}{2} - \epsilon]^2}$  }},
{ {x8}, { { {G[{p1, p2, p3}, {p1, p2, p3}], ε},
{-G[{k1, p1 + p2, p3}, {k1, p1 + p2, p3}], -1 +  $\frac{1}{2} (2 + 2 \epsilon)$  }},
{G[{k1, p1 + p2, p2, p3}, {k1, p1 + p2, p2, p3}],  $\frac{1}{2} + \frac{1}{2} (-2 - 2 \epsilon)$  },
{G[{k1, k2, p1 + p2, p3}, {k1, k2, p1 + p2, p3}],  $\frac{1}{2} + \frac{1}{2} (-2 - 2 \epsilon)$  }},
 $\frac{1}{64 \pi^3 \text{Gamma}[\frac{1}{2} - \epsilon]^2}$  } } }

```

```

58 (*2. get the explicit expressions for the representation*)
59 (*we can also get the dimension of the top topology under maximal cut*)
60 (*dim means the number of master integrals under maximal cut without considering
61 toprepexp = Times@@Power@@@toprep[[1,2,1]]//Gram2Poly[#,krep]& //Simplify
62 dim = GetDimension[toprepexp/.{xi_/:;(i<8)->0},{x8}]

```

```

Out[25]= 161+ε (-s t (s + t))ε (s (m2 s + x5 (x7 - x8) + (s - x7) x8 + x82))ε
(-4 m2 s2 t - 4 m2 s t2 + s2 t2 + t2 x52 + s2 x62 -
2 s t2 x7 + t2 x72 - 2 t x5 (-s x6 + (2 s + t) x7 + s (t - x8)) -
2 s2 t x8 + 2 s t x7 x8 + s2 x82 - 2 s x6 (s t - t x7 + (s + 2 t) x8))- $\frac{1}{2}$ -ε
(-4 m2 s2 x4 + s2 x42 + 4 m2 s x3 x7 - 2 s x3 x4 x7 - 4 m2 s x5 x7 + x32 x72 +
x22 (s (-4 m2 + s) + x52 - 2 s x7 + x72 - 2 x5 (s + x7)) + x12 (x5 - x8)2 - 4 m2 s x3 x8 -
2 s2 x4 x8 + 2 s x3 x4 x8 + 4 m2 s x5 x8 + 4 m2 s x7 x8 + 2 s x3 x7 x8 - 2 x32 x7 x8 - 4 m2 s x82 +
s2 x82 - 2 s x3 x82 + x32 x82 + 2 x2 (x3 (2 m2 s - x72 + (s - x5) x8 + x7 (s + x5 + x8)) + s
(-2 m2 x7 + x4 (-s + x5 + x7 - 2 x8) + 4 m2 x8 - s x8 + x7 x8 + x5 (-2 m2 - 2 x7 + x8))) +
2 x1 (s (x5 - x8) (2 m2 - x4 + x8) - x3 (2 m2 s + x5 (x7 - x8) + (2 s - x7) x8 + x82) +
x2 (2 m2 s - x52 + (s - x7) x8 + x5 (s + x7 + x8)))- $\frac{1}{2}$ -ε

```

```

Out[26]= 4

```

```
In[31]:= (*3. considering the symmetry of top sector by analyzing the polynomial in star
66 poly=stdrep[1,2,1]//Gram2Poly[#,krep]&;
67 symrule=PolySym[poly,Join[Subscript[x,#]&/@Range[$BvNum],{s,t,m2}],$BvNum]
```

```
Out[32]= {{Cycles[{{2, 9}, {6, 8}}], Cycles[{{1, 2}, {3, 9}, {5, 6}, {7, 8}, {10, 11}}],
Cycles[{{1, 2, 3, 9}, {5, 6, 7, 8}, {10, 11}}],
Cycles[{{1, 3}, {5, 7}}], Cycles[{{1, 3}, {2, 9}, {5, 7}, {6, 8}}],
Cycles[{{1, 5}, {2, 6}, {3, 7}, {8, 9}}],
Cycles[{{1, 5}, {2, 8}, {3, 7}, {6, 9}}],
Cycles[{{1, 6}, {2, 5}, {3, 8}, {7, 9}, {10, 11}}],
Cycles[{{1, 6, 3, 8}, {2, 7, 9, 5}, {10, 11}}],
Cycles[{{1, 7}, {2, 6}, {3, 5}, {8, 9}}],
Cycles[{{1, 7}, {2, 8}, {3, 5}, {6, 9}}],
Cycles[{{1, 8, 3, 6}, {2, 5, 9, 7}, {10, 11}}],
Cycles[{{1, 8}, {2, 7}, {3, 6}, {5, 9}, {10, 11}}],
Cycles[{{1, 9, 3, 2}, {5, 8, 7, 6}, {10, 11}}],
Cycles[{{1, 9}, {2, 3}, {5, 8}, {6, 7}, {10, 11}}]},
{{X2 → X9, X6 → X8, X8 → X6, X9 → X2},
{X1 → X2, X2 → X1, X3 → X9, X5 → X6, X6 → X5, X7 → X8, X8 → X7, X9 → X3, s → t, t → s},
{X1 → X9, X2 → X1, X3 → X2, X5 → X8, X6 → X5, X7 → X6, X8 → X7, X9 → X3, s → t, t → s},
{X1 → X3, X3 → X1, X5 → X7, X7 → X5},
{X1 → X3, X2 → X9, X3 → X1, X5 → X7, X6 → X8, X7 → X5, X8 → X6, X9 → X2},
{X1 → X5, X2 → X6, X3 → X7, X5 → X1, X6 → X2, X7 → X3, X8 → X9, X9 → X8},
{X1 → X5, X2 → X8, X3 → X7, X5 → X1, X6 → X9, X7 → X3, X8 → X2, X9 → X6},
{X1 → X6, X2 → X5, X3 → X8, X5 → X2, X6 → X1, X7 → X9, X8 → X3, X9 → X7, s → t, t → s},
{X1 → X8, X2 → X5, X3 → X6, X5 → X9, X6 → X1, X7 → X2, X8 → X3, X9 → X7, s → t, t → s},
{X1 → X7, X2 → X6, X3 → X5, X5 → X3, X6 → X2, X7 → X1, X8 → X9, X9 → X8},
{X1 → X7, X2 → X8, X3 → X5, X5 → X3, X6 → X9, X7 → X1, X8 → X2, X9 → X6},
{X1 → X6, X2 → X7, X3 → X8, X5 → X2, X6 → X3, X7 → X9, X8 → X1, X9 → X5, s → t, t → s},
{X1 → X8, X2 → X7, X3 → X6, X5 → X9, X6 → X3, X7 → X2, X8 → X1, X9 → X5, s → t, t → s},
{X1 → X2, X2 → X3, X3 → X9, X5 → X6, X6 → X7, X7 → X8, X8 → X5, X9 → X1, s → t, t → s},
{X1 → X9, X2 → X3, X3 → X2, X5 → X8, X6 → X7, X7 → X6, X8 → X5, X9 → X1, s → t, t → s}}}
```

```
In[33]:= (*check these are indeed the symmetry of this polynomial*)
71 PolySymCheck[poly,symrule[[2]]]
```

```
Out[33]= {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}
```

```

In[39]:= (*4. get all zero sectors derived from Baikov representation*)
75 zerolist=GetMatZeroSector[resultmat,$BvNum,{8,9}] (*the last list is where the
76 PadRight[Reverse[IntegerDigits[#,2]], $BvNum]&/@zerolist (*loop up these sector

```

```

Out[39]= {0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21,
22, 23, 24, 32, 33, 34, 35, 36, 37, 38, 39, 40, 48, 49, 50, 51, 52, 53, 54,
55, 56, 64, 65, 66, 67, 68, 69, 70, 71, 72, 80, 81, 82, 83, 84, 86, 88, 96,
97, 98, 99, 100, 101, 102, 103, 104, 112, 113, 114, 115, 116, 118, 120}

```

```

Out[40]= {{0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 1, 0, 0, 0, 0, 0, 0, 0, 0},
{1, 1, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 1, 0, 0, 0, 0, 0, 0, 0}, {1, 0, 1, 0, 0, 0, 0, 0, 0, 0},
{0, 1, 1, 0, 0, 0, 0, 0, 0, 0}, {1, 1, 1, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 1, 0, 0, 0, 0, 0, 0},
{1, 0, 0, 1, 0, 0, 0, 0, 0, 0}, {0, 1, 0, 1, 0, 0, 0, 0, 0, 0}, {1, 1, 0, 1, 0, 0, 0, 0, 0, 0},
{0, 0, 1, 1, 0, 0, 0, 0, 0, 0}, {1, 0, 1, 1, 0, 0, 0, 0, 0, 0}, {0, 1, 1, 1, 0, 0, 0, 0, 0, 0},
{1, 1, 1, 1, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0, 0, 0, 0, 0}, {1, 0, 0, 0, 1, 0, 0, 0, 0, 0},
{0, 1, 0, 0, 1, 0, 0, 0, 0, 0}, {1, 1, 0, 0, 1, 0, 0, 0, 0, 0}, {0, 0, 1, 0, 1, 0, 0, 0, 0, 0},
{1, 0, 1, 0, 1, 0, 0, 0, 0, 0}, {0, 1, 1, 0, 1, 0, 0, 0, 0, 0}, {1, 1, 1, 0, 1, 0, 0, 0, 0, 0},
{0, 0, 0, 1, 1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 1, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 1, 0, 0, 0, 0},
{0, 1, 0, 0, 0, 1, 0, 0, 0, 0}, {1, 1, 0, 0, 0, 1, 0, 0, 0, 0}, {0, 0, 1, 0, 0, 1, 0, 0, 0, 0},
{1, 0, 1, 0, 0, 1, 0, 0, 0, 0}, {0, 1, 1, 0, 0, 1, 0, 0, 0, 0}, {1, 1, 1, 0, 0, 1, 0, 0, 0, 0},
{0, 0, 0, 1, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 1, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 1, 1, 0, 0, 0},
{0, 1, 0, 0, 0, 1, 1, 0, 0, 0}, {1, 1, 0, 0, 0, 1, 1, 0, 0, 0}, {0, 0, 1, 0, 0, 1, 1, 0, 0, 0},
{1, 0, 0, 0, 0, 1, 1, 0, 0, 0}, {0, 1, 0, 0, 0, 1, 1, 0, 0, 0}, {1, 1, 1, 0, 0, 1, 1, 0, 0, 0},
{0, 0, 1, 0, 0, 1, 1, 0, 0, 0}, {1, 0, 1, 0, 0, 1, 1, 0, 0, 0}, {0, 1, 1, 0, 0, 1, 1, 0, 0, 0},
{1, 1, 1, 0, 0, 1, 1, 0, 0, 0}, {0, 0, 0, 1, 0, 1, 1, 0, 0, 0}, {0, 0, 0, 0, 1, 1, 1, 0, 0, 0},
{1, 0, 0, 0, 1, 1, 1, 0, 0, 0}, {0, 1, 0, 0, 1, 1, 1, 0, 0, 0}, {1, 1, 0, 0, 1, 1, 1, 0, 0, 0},
{0, 0, 1, 0, 1, 1, 1, 0, 0, 0}, {0, 1, 1, 0, 1, 1, 1, 0, 0, 0}, {0, 0, 0, 1, 1, 1, 1, 0, 0, 0}}

```

Linear propagator involved

```

In[56]:= (*now we present above example with linear propagator involved*)
84  (*just replace PD with LPD: LPD[k,p,-m2]=k.p-m2, PD[k,p,-m2]=(k+p)^2-m2*)
85  dlist={
86      PD[k1,0,-m2],
87      LPD[k1,-p1,-m2],
88      LPD[k1,-p1-p2,-m2],
89      LPD[k1,k2,0],
90      PD[k2,-p1-p2,-m2],
91      PD[k2,-p1-p2-p3,-m2],
92      PD[k2,0,-m2],
93      LPD[k2,-p1,-m2],
94      PD[k1,-p1-p2-p3,-m2]
95  };(*input of propagators and independent scalar products*)
96  conservation={p4→-p1-p2-p3};(*if dependent variables does not appear, this can
97  kinematics={
98      SProd[p1]→0,
99      SProd[p2]→0,
100     SProd[p3]→0,
101     SProd[p4]→0,
102     SProd[p1+p2]→s,
103     SProd[p2+p3]→t
104  };(*define independent scalar products, SProd[a]=SProd[a,a]*)

In[59]:= kinerep=GetIndExtSProd[kinematics,conservation];(*represent all scalar products
108  nkrep=Join[BaikovTrans[dlist,kinerep][[2]],kinerep];(*a replacement rule for all

```

```

111 (*one can compare the result with the former example*)
112 nkrep(*new one with linear propagator*)
113 krep(*former one with all normal propagators*)

```

```

Out[54]= {k1 · k1 → m2 + x1, k1 · k2 → x4, k1 · p1 → -m2 - x2, k1 · p2 → x2 - x3,
          k1 · p3 →  $\frac{1}{2} (2 m^2 + x_1 + 2 x_3 - x_9)$ , k2 · k2 → m2 + x7, k2 · p1 → -m2 - x8,
          k2 · p2 →  $\frac{1}{2} (2 m^2 + s - x_5 + x_7 + 2 x_8)$ , k2 · p3 →  $\frac{1}{2} (-s + x_5 - x_6)$ , p1 · p1 → 0,
          p1 · p2 →  $\frac{s}{2}$ , p1 · p3 →  $-\frac{s}{2} - \frac{t}{2}$ , p2 · p2 → 0, p2 · p3 →  $\frac{t}{2}$ , p3 · p3 → 0}

```

```

Out[55]= {k1 · k1 → m2 + x1, k1 · k2 →  $\frac{1}{2} (2 m^2 + x_1 - x_4 + x_7)$ , k1 · p1 →  $\frac{1}{2} (x_1 - x_2)$ ,
          k1 · p2 →  $\frac{1}{2} (s + x_2 - x_3)$ , k1 · p3 →  $\frac{1}{2} (-s + x_3 - x_9)$ , k2 · k2 → m2 + x7,
          k2 · p1 →  $\frac{1}{2} (x_7 - x_8)$ , k2 · p2 →  $\frac{1}{2} (s - x_5 + x_8)$ , k2 · p3 →  $\frac{1}{2} (-s + x_5 - x_6)$ ,
          p1 · p1 → 0, p1 · p2 →  $\frac{s}{2}$ , p1 · p3 →  $-\frac{s}{2} - \frac{t}{2}$ , p2 · p2 → 0, p2 · p3 →  $\frac{t}{2}$ , p3 · p3 → 0}

```

```

In[61]:= (*derive the representations in the same way*)
117 stdrep=BaikovRep[dlist,{k1,k2},{p1,p2,p3},Abst→True];(*the standard representation*)
118 nresultmat=AllSectorBaikovMat[stdrep,nkrep,Exc→{}];(*derive all representation*)

```

The global variable \$LoopNum has been set to 2

The global variable \$BvNum has been set to 9

additional parameters in definition of propagators

```
In[67]:= (*in some cases, there may be parameters between the combinations of momenta*)
126 (*we need to declare these variables as numbers*)
127 DeclareVarAsNum[{r,z}]
128 dlist={
129     PD[l1, -(1 - z)*k1, 0],
130     PD[l1, z*k1, 0],
131     PD[l2, 0, 0],
132     PD[l3, 0, 0],
133     PD[l1 - l2, -(1 - z)*k1, 0],
134     PD[l1 + l3, z*k1, 0],
135     PD[l1 + l3, z*k1 + k2, 0],
136     PD[l1 - l2 + l3, z*k1 + k2, 0],
137     PD[l1 - l2, z*k1 + k2, 0],
138     PD[l2, k1, 0],
139     PD[l3, k1, 0],
140     PD[l1, k2, 0]
141 };(*input of propagators and independent scalar products*)
142 conservation={};(*if dependent variables does not appear, this can be an empty
143 kinematics={
144     SProd[k1]→0,
145     SProd[k2]→0,
146     SProd[k1,k2]→s/2
147 };(*define independent scalar products, SProd[a]=SProd[a,a]*)
```

variables: {r, z} have been declared as numbers.

```
In[71]:= kinerep=GetIndExtSProd[kinematics,conservation];(*represent all scalar products
151 krep=Join[BaikovTrans[dlist,kinerep][[2]],kinerep];(*a replacement rule for all s
```

```
In[79]:= krep
```

```
Out[79]= {k1 · l1 →  $\frac{1}{2} (-x_1 + x_2)$ , k1 · l2 →  $\frac{1}{2} (-x_3 + x_{10})$ ,
k1 · l3 →  $\frac{1}{2} (-x_4 + x_{11})$ , k2 · l1 →  $\frac{1}{2} (-z x_1 - x_2 + z x_2 + x_{12})$ ,
k2 · l2 →  $\frac{1}{2} (s z - x_1 - z x_1 + z x_2 + x_3 + x_5 - x_9 - x_{10} + x_{12})$ ,
k2 · l3 →  $\frac{1}{2} (-s z + z x_1 + x_2 - z x_2 - x_6 + x_7 - x_{12})$ , l1 · l1 →  $z x_1 + x_2 - z x_2$ ,
l1 · l2 →  $\frac{1}{2} (x_1 + z x_3 - x_5 + x_{10} - z x_{10})$ , l1 · l3 →  $\frac{1}{2} (-x_2 - x_4 + z x_4 + x_6 - z x_{11})$ ,
l2 · l2 →  $x_3$ , l2 · l3 →  $\frac{1}{2} (-s z + z x_1 - z x_2 + x_7 - x_8 + x_9 - x_{12})$ ,
l3 · l3 →  $x_4$ , k1 · k1 → 0, k1 · k2 →  $\frac{s}{2}$ , k2 · k2 → 0}
```

```
In[80]:= stdrep=BaikovRep[dlist,{l1,l2,l3},{k1,k2},Abst→True];(*the standard representa-
158 resultmat=AllSectorBaikovMat[stdrep,krep];(*derive all representations from the
```

The global variable \$LoopNum has been set to 3
The global variable \$BvNum has been set to 12

```
In[82]:= Export[current<>"para_rep.wxf",resultmat,PerformanceGoal→"Size"];
```

```
In[84]:= Options[GetBaikovMatRep]
```

```
Out[84]= {ExcVar → {}, looporder → 3, ForceAdd → 0, Baikov`Private`debug → False}
```

```
In[85]:= GetBaikovMatRep[resultmat,{1,2,3,4,5,6,7,8,9}]/Simplify
```

```
Out[85]= {{ {x11},

$$\left\{ \left\{ G[\{k_1, k_2\}, \{k_1, k_2\}], -\frac{1}{2} + \epsilon \right\}, \{-G[\{l_1 + k_1 z, l_2, k_2\}, \{l_1 + k_1 z, l_2, k_2\}], \right.$$


$$\epsilon\}, \left\{ G[\{l_1 + k_1 z, l_2, k_1, k_2\}, \{l_1 + k_1 z, l_2, k_1, k_2\}], -\frac{1}{2} - \epsilon \right\},$$


$$\left\{ G[\{l_1 + k_1 z, l_2, l_3, k_2\}, \{l_1 + k_1 z, l_2, l_3, k_2\}], -\frac{1}{2} - \epsilon \right\} \right\},$$


$$\frac{2^{-2(4+\epsilon)}}{\pi^{9/2} \Gamma[1-2\epsilon] \Gamma[\frac{1}{2}-\epsilon]} \left\{ \{x_{10}\}, \left\{ \left\{ G[\{k_1, k_2\}, \{k_1, k_2\}], -\frac{1}{2} + \epsilon \right\}, \right.$$


$$\{-G[\{2k_2 + l_1 - 2k_2 z, l_3, k_1 + k_2\}, \{2k_2 + l_1 - 2k_2 z, l_3, k_1 + k_2\}], \epsilon\},$$


$$\left\{ G[\{2k_2 + l_1 - 2k_2 z, l_3, k_1 + k_2, k_2\}, \{2k_2 + l_1 - 2k_2 z, l_3, k_1 + k_2, k_2\}], -\frac{1}{2} - \right.$$


$$\epsilon\}, \left\{ G[\{2k_2 + l_1 - 2k_2 z, l_2, l_3, k_1 + k_2\}, \{2k_2 + l_1 - 2k_2 z, l_2, l_3, k_1 + k_2\}], \right.$$


$$\left. -\frac{1}{2} - \epsilon \right\} \right\}, \frac{2^{-2(4+\epsilon)}}{\pi^{9/2} \Gamma[1-2\epsilon] \Gamma[\frac{1}{2}-\epsilon]} \left\{ \right\} \left\{ \right\} \left\{ \right\}$$

```


five-loop banana: Derive representation starting from somewhere not be the standard representation

```

In[86]:= dlist= {
175     PD[k1,0,-m2],
176     PD[k2-k1,0,0],
177     PD[k3-k2,0,0],
178     PD[k4-k3,0,0],
179     PD[k5-k4,0,0],
180     PD[k5,p,-m2],
181     PD[k2,0,-m2],
182     PD[k3,0,-m2],
183     PD[k4,0,-m2],
184     PD[k5,0,-m2],
185     PD[k1,p,-m2],
186     PD[k2,p,-m2],
187     PD[k3,p,-m2],
188     PD[k4,p,-m2],
189     PD[k5-k1,0,0],
190     PD[k5-k2,0,0],
191     PD[k5-k3,0,0],
192     PD[k4-k1,0,0],
193     PD[k4-k2,0,0],
194     PD[k3-k1,0,0]
195 };(*input of propagators and independent scalar products*)
196 conservation={};(*if dependent variables does not appear, this can be an empty
197 kinematics={
198     SProd[p]→s
199 };(*define independent scalar products, SProd[a]=SProd[a,a]*)

In[89]:= kinerep=GetIndExtSProd[kinematics,conservation];(*represent all scalar products
203 krep=Join[BaikovTrans[dlist,kinerep][[2]],kinerep];(*a replacement rule for all s

In[92]:= (*we use Exc→{...} to first specify which variables are not integrated out.*)
207 (*Then we get all possible loop-by-loop representations for the top sector*)
208 (*at last, we start from these loop-by-loop representations and derive all the
209 AbsoluteTiming[
210 stdrep=BaikovRep[dlist,{k1,k2,k3,k4,k5},{p},Abst→True];
211 resultmat=AllSectorBaikovMat[stdrep,krep,Exc→{x1,x2,x3,x4,x5,x6}];
212 resultc=AllSectorBaikovMatC[resultmat,Subscript[x,#]&/@Range[20],krep];
213 Export[current<>"banana5l_rep.wxf",resultc,PerformanceGoal→"Size"];
214 ]

```

The global variable \$LoopNum has been set to 5

The global variable \$BvNum has been set to 20

Out[92]= {319.26, Null}

```

In[105]:= (*there are totally 42 representations for this single top topology*)
218      (*but they are symmetric as we will see*)
219      toprep=GetBaikovMatRep[resultc,{1,2,3,4,5,6}];
220      toprep//Length

```

Out[106]=

42

```

In[101]:= (*half in integer power, half in half-integer power*)
224      toprep[[1]]//Simplify
225      toprep[[-1]]//Simplify

```

Out[101]=

$$\begin{aligned}
& \{ \{x_{20}, x_{19}, x_{18}, x_{17}, x_{16}, x_{15}, x_{14}, x_{13}, x_{12}, x_{11}\}, \\
& \left\{ \left\{ G[\{p\}, \{p\}], -1 + \epsilon, \{-G[\{k_2\}, \{k_2\}], -1 + \epsilon, \left\{ G[\{k_1, k_2\}, \{k_1, k_2\}], \frac{1}{2} - \epsilon \right\}, \right. \right. \\
& \quad \left. \left\{ -G[\{k_3\}, \{k_3\}], -1 + \epsilon, \left\{ G[\{k_2, k_3\}, \{k_2, k_3\}], \frac{1}{2} - \epsilon \right\}, \right. \right. \\
& \quad \left. \left\{ -G[\{k_4\}, \{k_4\}], -1 + \epsilon, \left\{ G[\{k_3, k_4\}, \{k_3, k_4\}], \frac{1}{2} - \epsilon \right\}, \right. \right. \\
& \quad \left. \left\{ -G[\{k_5\}, \{k_5\}], -1 + \epsilon, \left\{ G[\{k_5, p\}, \{k_5, p\}], \frac{1}{2} - \epsilon \right\}, \right. \right. \\
& \quad \left. \left. \left\{ G[\{k_4, k_5\}, \{k_4, k_5\}], \frac{1}{2} - \epsilon \right\} \right\}, \frac{(-1)^{-2\epsilon}}{32 \pi^{5/2} \Gamma\left[\frac{3}{2} - \epsilon\right]^5} \right\} \}
\end{aligned}$$

Out[102]=

$$\begin{aligned}
& \{ \{x_{19}, x_{17}, x_{16}, x_{14}, x_{13}, x_{12}, x_{10}, x_9, x_8, x_7\}, \\
& \left\{ \left\{ G[\{p\}, \{p\}], -1 + \epsilon, \{-G[\{k_1 - k_3\}, \{k_1 - k_3\}], -1 + \epsilon, \right. \right. \\
& \quad \left. \left\{ G[\{k_1 - k_3, k_2 - k_3\}, \{k_1 - k_3, k_2 - k_3\}], \frac{1}{2} - \epsilon \right\}, \right. \\
& \quad \left. \left\{ -G[\{k_1 - k_4\}, \{k_1 - k_4\}], -1 + \epsilon, \left\{ G[\{k_1 - k_4, k_3 - k_4\}, \{k_1 - k_4, k_3 - k_4\}], \frac{1}{2} - \epsilon \right\}, \right. \right. \\
& \quad \left. \left\{ -G[\{k_1 - k_5\}, \{k_1 - k_5\}], -1 + \epsilon, \left\{ G[\{k_1 - k_5, k_4 - k_5\}, \{k_1 - k_5, k_4 - k_5\}], \frac{1}{2} - \epsilon \right\}, \right. \right. \\
& \quad \left. \left\{ -G[\{k_1 + p\}, \{k_1 + p\}], -1 + \epsilon, \left\{ G[\{k_1 + p, p\}, \{k_1 + p, p\}], \frac{1}{2} - \epsilon \right\}, \right. \right. \\
& \quad \left. \left. \left\{ G[\{k_1 + p, k_5 + p\}, \{k_1 + p, k_5 + p\}], \frac{1}{2} - \epsilon \right\} \right\}, \frac{(-1)^{-2\epsilon}}{32 \pi^{5/2} \Gamma\left[\frac{3}{2} - \epsilon\right]^5} \right\} \}
\end{aligned}$$